

Instructor: Date:

Learner: Regional Branch:

Program: Center:

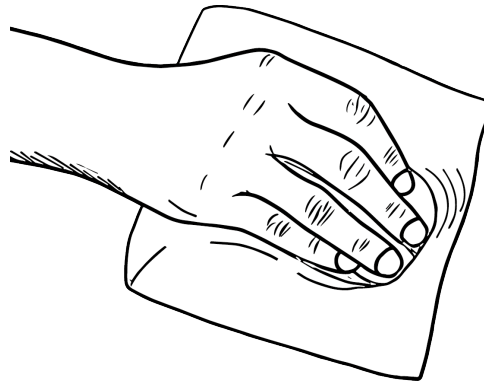
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Applications are event-driven on ASP.NET.



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can say what an event is.
- I can talk about important application events in the ASP.NET.
- I can handle simple events using controls in ASP.NET.

Ask your partner:

- How many sections can you find in an ASP.NET page? Why are these sections different?
- What do you think the processes involved when running a web page or application are?





Let's Explore!

Gloria has been beating around the bush* with her presentation on event handling.

*To "beat around the bush" is when someone says or does something without getting to the point.



Instructions: Read the text below.

- Javier:** What's up, Gloria? You look worried.
- Gloria:** I'm stressing out. Tomorrow I have a presentation about events in ASP.NET. I've been beating around the bush and I have nothing yet.
- Javier:** Don't worry. I'm an expert on that matter! As you know, ASP.NET offers an event-driven way of building pages or applications. It means that every action or request that is activated by the user is an event. You may face an event, for example, when you press a key or click on a tab when exploring a web page.
- Gloria:** So, every action that generates a notification within the system of a web page can be considered an event. Right?
- Javier:** Exactly. All procedures in ASP.NET communicate via events; in this way, events aren't designed only to respond to actions but also to manage them.
- Gloria:** Could you explain to me how the event handling process is? It is not clear to me if requests are handled or not by servers.
- Javier:** Well, events are not activated by the server machine, but by the client's. For instance, if you click on a button, your request raises a click event. Then, the browser sends it to the server where there is a subroutine or event handler that explains what to do in such a case; and finally, this event handler is executed to send a response back to the client machine.
- Gloria:** It's easier to understand if you put it that way. Do you know how events are defined? It is not well explained in the articles that I've been reading.
- Javier:** When you set an event handler, its syntax normally needs to include an argument or parameter in the first place, and return voids or values in the other.
- Gloria:** I see. And what types of events may I need to build an application in ASP.NET?
- Javier:** Well, we could say that the most important application events in the ASP.NET are `Application_Start` and `Application_End`. The first one is raised when the user initiates the application and the second one when the user ends it.
- Gloria:** OK. How do I set event handlers when using controls?
- Javier:** Events aren't generated out of the blue; a user needs to perform an action in a controller to trigger the event. Control events are classified depending on input actions; for instance, click or command events are connected with buttons and text changed events are related to text boxes.



Let's Practice!

ASP.NET offers an event-driven way of building pages or applications.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. Events are not considered simple actions;
2. Actions such as clicking on a tab aren't called results,
3. Events aren't raised at the server machine.
4. Events in ASP.NET aren't managed by the user,
5. Events aren't generated out of the blue;

COLUMN B

- a. They are activated at the client machine.
- b. a user needs to perform an action in a controller to trigger the event.
- c. they are called events.
- d. notifications are generated within the system of a web page every time a user sends a request.
- e. they are managed by the application.



Let's Practice!

What types of events may I need to build an application in ASP.NET?

Instructions: Use the words from the box to correctly complete the sentences.

needs to include / initiates / raises / stops / sends

1. If you click on a button, your request _____ a click event.
2. The browser _____ the request to the server where there is a subroutine or event handler that explains what to do in such case.
3. The event syntax normally _____ an argument or parameter in the first place
4. When the user _____, the application event raised is Application_Start.
5. When the application _____, the application event raised is Application_End.





Extension Activity

Discovering Control Events in ASP.NET.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search more information about control events in ASP.NET.
- Identify the most common control events, their attributes and functionalities.
- Create a chart where you explain control events in ASP.NET.
- Add examples in your chart to illustrate how each event works.
- Prepare an oral presentation to explain your chart.
- Share your work with the class.

Self-Assessment

Yes

No

- I search more information about control events in ASP.NET. ☐ Yes ☐ No
- I identify the most common control events, their attributes and functionalities. ☐ Yes ☐ No
- I create a chart to explain control events in ASP.NET. ☐ Yes ☐ No
- I prepare an oral presentation to explain my chart. ☐ Yes ☐ No
- I share my work with the class. ☐ Yes ☐ No



Target Words

- argument** (noun): a variable given to a section of data in a subroutine
e.g. When you set an event handler, its syntax normally needs to include an argument or parameter in the first place, and return voids or values in the other.
- event-driven** (adjective): based on or operated by events or actions.
e.g. ASP.NET offers an event-driven way of building pages or applications.
- subroutine** (noun): a set of steps to perform a specific task
e.g. Then, the browser sends it to the server where there is a subroutine or event handler that explains what to do in such a case.
- return void** (noun): a function that does not return any value
e.g. When you set an event handler, its syntax normally needs to include an argument or parameter in the first place, and return voids or values in the other.
- trigger** (verb): to activate something
e.g. Events aren't generated out of the blue; a user needs to perform an action in a controller to trigger the event.
- tab** (noun): a part of a website that gives access to something
e.g. You may face an event, for example, when you press a key or click on a tab when exploring a web page.



Learning Points

Simple Present Passive in Negative Form.

You use **simple present passive** in **negative** statements to express that something doesn't happen frequently.

When we use a negative statement in active voice, we aim to highlight the person or element that performs or not an action:

- Servers **do not** post back some events immediately.

But when it comes to highlighting the role of the object in a sentence, we need to use the passive voice. When we **use simple present passive** in **negative** statements, we often express that something is not done as it is supposed to be done on a regular basis:

- Some events **are not posted** back immediately.
- Some events **are not posted** back by servers immediately.

When using the passive voice the subject could be omitted, but in case you wish to mention it, you need to use the particle **by** followed by the noun.

This is the grammatical structure we need to use when building negative statements in **simple present passive**:

Object + **Verb Be in Negative Form** + **Verb in Past Participle** + Complement

Example:

- The event **is not described** until the server has a subroutine.

Examples

- Events aren't designed only to respond to actions but also to manage them.
- It is not well explained in the articles that I've been reading.
- Events aren't generated out of the blue; a user needs to perform an action in a controller to trigger the event.





Learning Points

Common Control Events.

These are some common page and control events for ASP.NET.

In the following chart you will be able to identify some common events that appear when working with ASP.NET.

